

Jack, Yokes and Stem

A key function is the need to lower the slabs by hydraulic jacks. A modular system has been developed to accompany the chosen hydraulic jack type. The components are summarised below.

1 Hydraulic Jack

A number of different jacking systems were considered. A Bar jack was ultimately chosen as this worked best in the required headroom and allowed all access and operatives to be top side – a key safety consideration.

The Proposed Jack is the Following:

Jack Model	CRI-30/260-FO
Jack Manufacturer	FPT Fluid Power Technology
Jack Stroke	260mm
Capacity	300kN (push)

2 Jack Yokes

Given the expected cases in the wider project, two types of yoke have been developed. A single jack yoke and a double jack yoke. Both yokes share the same bolting pattern for interchangeability on stems.

The height between the top and bottom flanges of the yoke units allow for a Dywidag hexagonal nut to be engaged at the end of each jack stroke.

3 Spreaders

To provide the required punching shear capacity at the jacking points, a spreader assembly has been developed. This assembly increases the effective bearing area, and being a welded unit, can be retrieved and reused.

4 Jack Stem

To rationalise the temporary works, the jacking yokes will be bolted to a temporary steel universal column (UC). This approach avoids fixing the jacking units directly to the existing RC columns, which would otherwise generate high prying forces, require multiple anchors, and increase the risk of clashes with column reinforcement.

The temporary UC distributes the jacking forces back into the RC column, significantly reducing the anchor demand to a level where only two anchors are required.

Stem Installation Tie

The Lower stem will need to be fixed to the column during installation to prevent toppling until the upper stem is attached. This fixing should remain in place to allow the stem to be removed in the reverse sequence.

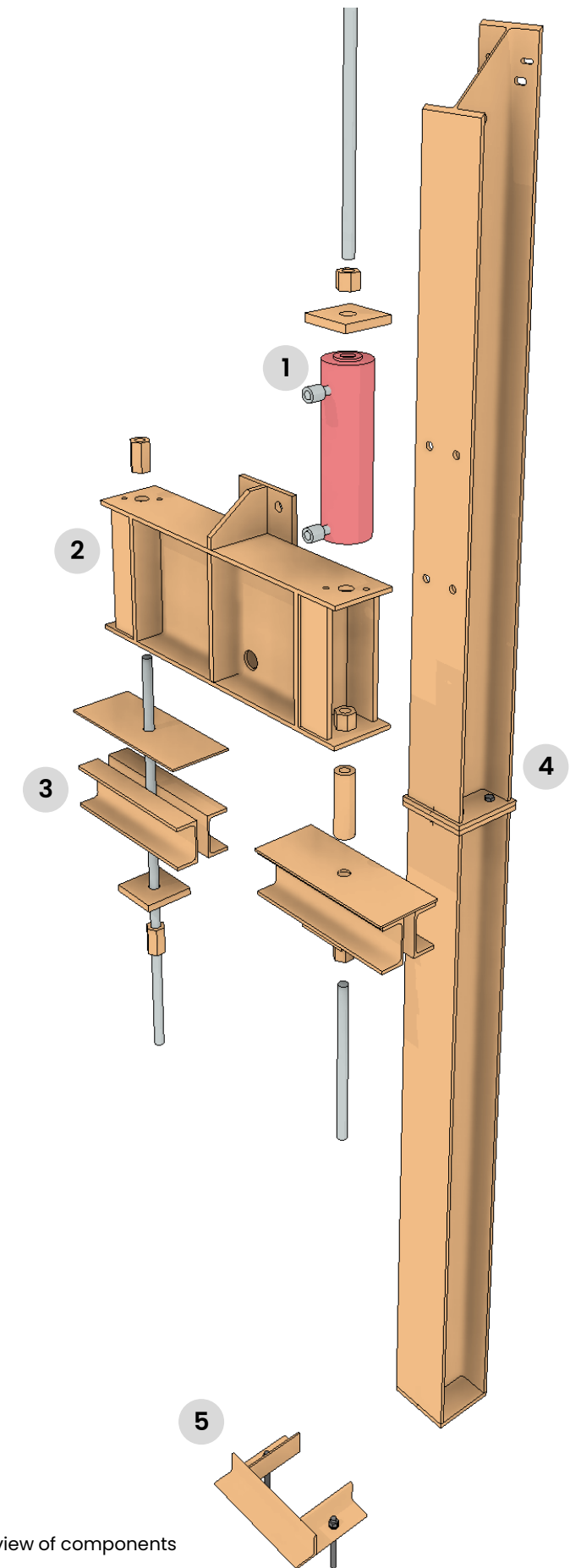
Stem Top Ties

The stem will need to be tied back to the adjacent permanent column at the top.

5 Stem Shoe

The Jack stem will be founded on the existing pile cap without anchors – this is to allow the stem to be removed later. To secure the stem at the base a shoe will be fixed to the pile cap using mechanical anchors. This will provide a horizontal restraint to the stem but allow it to be removed vertically.

The Stem shoe is the only part that will be sacrificial to the jacking process, all other components will be removed.



Exploded view of components